MILITARY SPECIFICATION

CARTRIDGES, DELAY, MARK 3 MOD 0 AND MARK 86 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 3 MOD 0 and MARK 86 MOD 0 delay cartridges and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-223	Powder, Black
MIL-P-18613	Primer, MARK 102 MOD 1
MIL-M-21383	Manganese Delay Composition
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-22264	Powders, Ignition, Gasless

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS AND ASSOCIATED LISTS

Naval Air Systems Command (Code Ident No. 30003)

LD 299325	Cartridge, Delay MARK 3 MOD 0
LD 419689	Container, Inner for Cartridges, Power Source
LD 419694	Test Set, Delay Cartridge, MARK 309 MOD 1
LD 491835	Container, Outer for Cartridges, Power Source
LD 534545	Cartridge, Delay, MARK 86 MOD 0
1298686	Cartridge, Delay, MARK 3 MOD 0, Assembly
1298687	Delay Body, Loading
1669377	Container, Inner, for Cartridges, Power Source, Assembly
1863082	Container, Outer, for Cartridges, Power Source, Assembly
2164754	Cartridge, Delay, MARK 86 MOD 0, Assembly

Naval Ordnance Systems Command (Code Ident No. 10001)

LD 275909	Primer, MARK 102 MOD 1

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PUBLICATIONS

Bureau of Naval Weapons

WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publication required by contractors in connection with specific procurement functic should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Department of Transportation, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintende of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General. The MARK 3 MOD 0 and the MARK 86 MOD 0 delay cartridges are identical except that the rim of the MARK 86 MOD 0 is nominally 0.087 inches thinner than the rim of the MARK 3 MOD 0.

3.1.1 The MARK 3 MOD 0 and the MARK 86 MOD 0 delay cartridges covered by this specification shall be manufactured in accordance with LD 299325 and LD 534545, respectively, and all documents listed thereon

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referenced on th respective NAVAIR drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy field activity prior to manufacture. When alternate materials or methods of cartridge manufacture are specified on the drawings, the bidder's selec tions shall be clearly stated in the proposal.

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3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 94 MARK 3 MOD 0 or MARK 86 MOD 0 delay cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purpose of this specification the Primer, MARK 102 MOD 1, LD 275909; the Manganese Delay Composition, MIL-M-21383; and Powders, Ignition, Gasless, MIL-P-22264 are defined as primary components.

3.4 Primer. The primers shall be furnished from lots manufactured within the preceding 12 months. The primers shall be manufactured in accordance with LD 275909 and shall be from lots which were accepted in accordance with MIL-P-18613. See 4.7.1.

3.4.1 X-ray. Primers shall be free of the following defects when examined by X-ray in accordance with 4.7.2: missing or abnormally thin charge; charge broken, cracked or porous; and all other observable defects in assembly.

3.5 Manganese delay composition. The manganese delay composition shall be manufactured in accordance with MIL-M-21383 and loaded in accordance with Drawing 1298687.

3.6 Gasless ignition powders. The gasless ignition powder A-1A shall be manufactured in accordance with MIL-P-22264, Class A and shall be loaded in accordance with Drawing 1298687.

3.7 Propellant charge.

3.7.1 Quantity. All propellant charges shall be as specified on Drawing 1298686 or Drawing 2164754, whichever applies. Charges shall be determined by weight rather than by volume measurements.

3.7.2 Black powder. The moisture content of the black powder shall be determined as specified in MIL-P-223 and at the time of loading shall be within the limits specified therein for the applicable class of black powder being used. See 4.7.3.

3.8 Cartridge inspection.

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3.8.1 Visual inspection. Cartridges shall be free of the following visible defects: dents, deep scratches, sharp edges, cracks, burrs, split or cracked lip in the crimped area, damaged closure disc, incomplete or improper sealant application, or any other imperfections which could prevent entry into the firing chamber, or adversely affect performance or resistance to moisture. See 4.7.4.

3.8.2 X-ray. Cartridges shall be free of imperfections when examined by X-ray in accordance with 4.7.5.

3.8.2.1 Serial number identification. Each cartridge in the production lot shall be legibly identified by serial numbers in accordance with 4.7.5 and the X-ray plates of the entire production lot shall accompany the ballistic test sample to the activity conducting the production lot acceptance test (see 6.2).

3.9 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.8.1, shall meet the requirements listed below when subjected to the test prescribed in 4.8.1.

3.9.1 Delay time. The lower and upper limits of delay time shall be L = 0.70 second and U = 1.20 seconds respectively over the temperature range from -65° $^{+0}_{-5}$°F to +160° $^{+5}_{-0}$°F.

3.9.2 Output. The lower and upper limits of the output shall be L = 70 foot-pounds and U = 185 foot-pounds over the temperature range from -65° $^{+0}_{-5}$°F to +160° $^{+5}_{-0}$°F.

3.9.3 Misfire. There shall be no misfires. See 4.8.2.3.

3.10 Periodic tests. The first five production lots from a single supplier shall pass the environmental tests of 4.9.1 and 4.9.2 after which only cartridges from each fifth production lot shall be subjected to these tests. If any lot fails to pass these tests, they shall be required on every lot until five consecutive lots have passed.

3.11 Packaging, Packing and Marking. The packaging and packing of cartridges and the container markings shall be in accordance with Section 5.

3.12 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 94 MARK 3 MOD 0 or MARK 86 MOD 0 cartridges, whichever applies, shall be prepared. Of these cartridges, 84 shall be expended in the tests listed below and 10 shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.4 through 3.8.2 of this specification. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-21625: 40 foot drop test (6), 14 day temperature and humidity test (18), vibration test (12), 15 "g" shock test (12). Number in parentheses denotes number of cartridges allocated for test. These cartridges (with the exception of the 40 foot drop cartridges), after exposure to the environmental treatments, together with 12 untreated cartridges shall be temperature conditioned for a minimum of six hours at 70° ± 5°F and fired in the test set of 4.8.1. In addition 24 cartridges shall be temperature conditioned for a minimum of six hours (12 at -65° ± 5°F and 12 at +160 ± 5°F) and fired in the test set of 4.8.1. All cartridges shall fire and produce results within the limits set forth in 3.9.1 and 3.9.2. The failure of any cartridge to meet the acceptance requirements as stated above shall cause rejection of the preproduction sample. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk.

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4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 501 and 3200 cartridges, respectively. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Quality conformance. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of LD 299325 or LD 534545, whichever applies, and verification of the following requirements and tests:

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Item	Nature of Requirement Verification	Applicable Paragraphs	Classification (WR-43)
Primary component	Single lot	3.3; 4.3.3	(M101)
Primer	Verification	3.4; 4.7.1	(C1)
Primer	X-ray	3.4.1; 4.7.2	(C2)
Manganese Delay Composition	Acceptance and loading	3.5	(C3)
Gasless ignition powder	Acceptance and loading	3.6	(C4)
Propellant	Type and weight	3.7.1	(C5)
Black powder	Condition	3.7.2; 4.7.3	(C6)
Cartridge	Visual inspection	3.8.1; 4.7.4	(M102)
Cartridge	X-ray	3.8.2; 4.7.5	(C7)
Cartridge	Serial number identification	3.8.2.1; 4.7.5	(M103)
Cartridge	Ballistic test	3.9; 4.8	(C8)
Cartridge	Packaging, packing and container markings	3.11; 4.6.1	(M104)

4.6.1 Packaging, packing and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Primer, verification. Verification shall be provided that the primers to be used were manufactured, loaded and accepted in accordance with the requirements of 3.4 and are from lots manufactured within the preceding 12 months.

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4.7.2 Primer X-ray. Examine all primers by X-ray and reject those having any of the defects of 3.4.1. For the most revealing exposure the primers are laid on their side. Pressure sensitive tape may be used as an aid in positioning the primers.

4.7.3 Moisture content of black powder. Conduct at least one test for each cartridge lot (or daily if this requires more frequent testing) and such additional tests as considered warranted by variations in conditions or process. Samples may be collected at the site of loading from the powder container, or taken from cartridges prior to sealing the black powder charge in the cartridges. The presence of moisture in excess of the limitations of MIL-P-223 shall cause suspension of loading until corrective measures have been taken.

4.7.4 Visual inspection. Examine all completely assembled cartridges and reject those having any of the defects noted in 3.8.1.

4.7.5 X-ray examination. Examine all cartridges by X-ray and reject those having any imperfections. All cartridges shall be identified with serial numbers prior to X-ray examination. The cartridge serial numbers shall be in consecutive order beginning with the number one (1) in each production lot. The cartridges for X-ray shall be arranged on trays or boards in consecutive numerical order and each X-ray plate shall carry a permanent identification of the cartridges displayed thereon. The X-ray plate identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. The serial numbers shall be rubber stamped on the cartridges with approximately 1/8 inch characters using the same process as that applying to the other cartridge markings. These serial numbers should be located as far apart as possible from the other cartridge markings.

4.8 Ballistic tests.

4.8.1 The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of the applicable test listed in 4.8.1.1 through 4.8.1.3. The cartridges shall be conditioned for a minimum of 6 hours at the temperature prescribed in 4.8.1.1 through 4.8.1.3 and fired in the MARK 309 MOD 1 delay cartridge test set, LD 419694. In addition to the samples of cartridges for the above tests, 10 additional cartridges shall be furnished from the production lot for investigative purposes.

4.8.1.1 Normal temperature. Cartridges shall be selected in accordance with MIL-STD-105, Table I, Inspection Level II. The cartridges shall be conditioned at a temperature of 70° ± 5°F for the designated time (see 4.8.1) prior to testing.

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4.8.1.2 Low temperature. Cartridges shall be selected in accordance with MIL-STD-105, Table I, Inspection Level S-3. The cartridges shall be conditioned at a temperature of -65° $^{+0}_{-5}$°F for the designated time (see 4.8.1) prior to testing.

4.8.1.3 High temperature. Cartridges shall be selected in accordance with MIL-STD-105, Table I, Inspection Level S-3. The cartridges shall be conditioned at a temperature of +160° $^{+5}_{-0}$°F for the designated time (see 4.8.1) prior to testing.

4.8.2 Criteria of acceptability.

4.8.2.1 Delay time. The acceptance number shall be zero and the rejection number shall be one for all tests, using the limits prescribed in 3.9.1.

4.8.2.2 Output. The acceptance number shall be zero and the rejection number shall be one for all tests, using the limits prescribed in 3.9.2.

4.8.2.3 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.9 Periodic tests. The number of cartridges for each test (see 3.10) shall be in accordance with Table I, Inspection Level S-3, MIL-STD-105. A separate sample is required for each test.

4.9.1 Temperature and humidity. The sample cartridges shall be subjected to the 14-day temperature and humidity cycling test of MIL-D-21625. The cartridges shall then be further subjected to the ballistic test of 4.8.1.1. All test cartridges shall satisfy the acceptance criteria of 4.8.2.

4.9.2 Vibration. The sample cartridges shall be subjected to the vibration tests of MIL-D-21625. The cartridges shall then be further subjected to the ballistic test of 4.8.1.1. All test cartridges shall satisfy the acceptance criteria of 4.8.2.

4.10 Retest. There shall be no retests.

4.10.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and that part of the test reconducted.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges of the same MARK and MOD shall be packaged in an inner container in accordance with LD 419689.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with LD 491835.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, cartridges intended for expenditure in preproduction sample testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on Drawing 1669377 and Drawing 1863082, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

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6. NOTES

6.1 Intended use. The MARK 3 MOD 0 delay cartridge was originally designed for use in the Belt, Aircraft Pilot's Safety, Automatic Release Cartridge Actuated MS 16036. The MARK 86 MOD 0 delay cartridge was originally designed for use in the harness release actuator, Douglas Aircraft Corporation P/N 3678132.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. directions for shipping x-ray plates of the entire production lot along with the ballistic test sample (see 3.8.2.1) to the activity conducting the production lot acceptance test.

e. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command (AIR-53222) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. For the purpose of this specification the following definitions apply.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

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6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirement of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirement of this level.

6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in MIL-STD-1314.

6.5.1 Applicability. The loading, assembly and handling of the explosives, explosive-loaded subassemblies and the finished items covered by this specification involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed in MIL-STD-1314 are mandatory. These same minimum provisions are recommended for observance by a contractor or manufacturer that explosively loads or assembles the item covered in this specification in order to fulfill his own safety responsibility.

Project No. 1377-N408

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-21048D(AS) CARTRIDGES, DELAY, MARK #3 MOD 0 and MK 86 MOD 0

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
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